

Market trends and projections

The market for TIMs is rapidly expanding across various industries, driven by their applications in electric vehicles (EVs), data centres, 5G infrastructure, and advanced driver assistance systems. Here is an overview of the market size projections by industry from 2022 to 2034:

- **EV power electronics.** Although there is a shift towards lower thermal conductivity in EV batteries, multifunctional TIMs are still essential for providing adhesion, mechanical strength, and other critical properties.
- **Data centres.** The demand for TIMs in data centres is expected to rise significantly due to the growing needs of artificial intelligence and high-performance computing. These applications require materials with higher thermal conductivity to manage increased heat output.
- **5G infrastructure.** The 5G market necessitates TIMs with thermal conductivity typically exceeding 5W/mK. This higher thermal conductivity is crucial for managing the substantial heat output generated by densely packed network equipment.

managing thermal issues becomes increasingly complex. One significant challenge is the effectiveness of EMI shielding tools. With the rise in frequencies, traditional board-level shields become less effective as smaller wavelengths facilitate energy leakage through gaps, exacerbating issues like cavity resonance. Additionally, the high frequency and short wavelength of 5G signals pose difficulties in heat dissipation, requiring the development of advanced thermal solutions to manage the increased thermal loads effectively.

TIMs are essential in 5G infrastructure and play a pivotal role in consumer electronics, such as smartphones, laptops, and tablets. The trend towards smaller, more powerful devices has increased the need for efficient thermal management solutions.

Smartphones. The iPhone 15 Pro Max, released in 2023, has a TIM area of approximately 346mm² for the board and 1000mm² for the display. Graphite sheets are commonly used as heat spreaders due to their high in-plane thermal conductivity and thin application, though their lack of compressibility can be a challenge.

Graphite sheets. Graphite sheets are favoured in smartphones for their anisotropic thermal

conductivity, offering excellent in-plane thermal performance. However, they often require additional thermal pastes or pads to ensure proper contact and typically include a polymer or adhesive layer to create an electrically insulating barrier.

The evolution of TIMs is crucial for the future of electronics, particularly in the realms of 5G and consumer devices. As the industry moves towards higher frequencies and greater device densities, advanced thermal management solutions will be essential to maintain performance and reliability. Detailed reports on thermal management across various industries are available for those interested in a deeper dive into these technologies and market trends. The ongoing advancements in TIM technology will continue to play a vital role in the performance and development of electronic systems, ensuring that they operate efficiently and within desired temperature ranges. As we look to the future, the importance of TIMs in enabling the next generation of electronic devices cannot be overstated. **EFY**

This article is based on the webinar 'Comprehensive Overview of Thermal Interface Materials: Trends and Developments.' Akanksha Sondhi Gaur, a Senior Technology Journalist at EFY, transcribed and curated the content.

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